Trace Tables

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected loop-and-trace record.

List range values, trace a fresh run, repair the reset and build the independent repeated addition before reviewing invalid rounds.

Keep one habit: Keep code version, supplied values, initial state, updates and observed output together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Branching Logic

Precision in Algorithms

MAIN LESSON · SECTION 12

W1 · Explain included start and excluded stop

> Use the supplied main accumulator script.

Your evidence record

- List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

```
total = 0
for turn in range(1, 4):
    total = total + turn
print(total)
```

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the changed range before running

> Record turn and total before/after each addition.

Your evidence record

- Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Locate the accumulator reset

> Use the supplied faulty script and preserve its failed run.

Your evidence record

- For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

```
total = 0
for turn in range(1, 4):
    total = 0
    total = total + turn
print(total)
```

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Build a repeated addition independently

> Keep one first attempt before hints or feedback.

Read and explain independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Build a repeated addition independently

MAIN LESSON · SECTION 16

W5 · Inspect negative rounds after the independent task

> Separate language behaviour from the stated nonnegative-input contract.

Your evidence record

- In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

My reasoning and evidence

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week’s change to your existing project record.

Your evidence record

Trace each book’s genre, length, difficulty, and total score before sorting.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: **Trace** before running for **Loop**s and **trace table**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation.
A student evaluates the condition using the initial value. Locate the divergence.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Set score=2; add 3; if score>4 subtract 1; double score.

Trace every value.

Show your **trace**, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Set score to 2. Add 3, then double the result. Record the score after each operation. What changes if you double before adding?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 4 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation.
A student evaluates the condition using the initial value. Locate the divergence.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional reading-token jar starts with one token and gains three per round.

Python: total=1; for turn in range(4): total=total+3; after the **loop** print(total).

The update is inside the **loop**; print is outside.

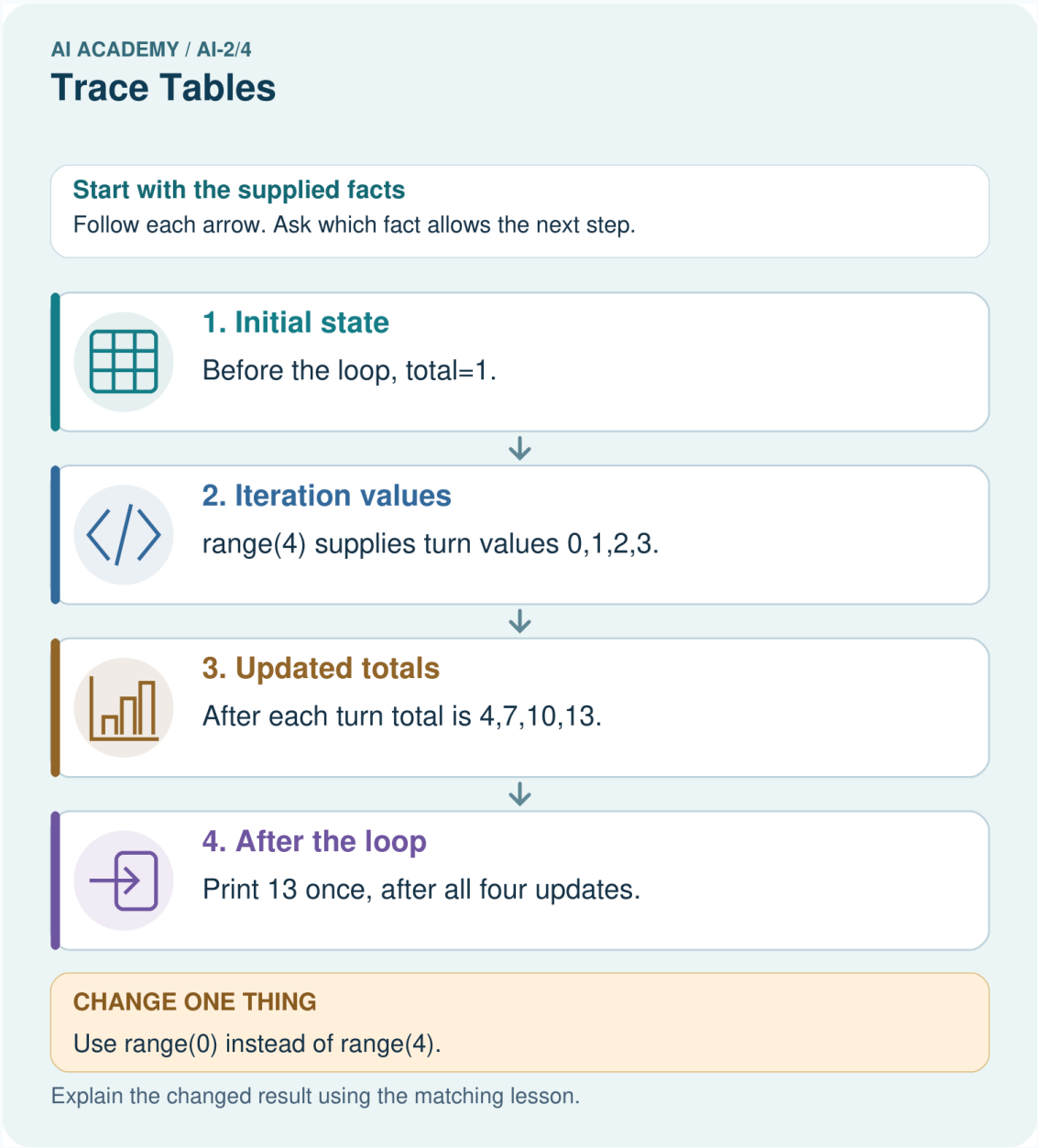
AI ACADEMY / AI-2/4

Loop totals after each iteration

turn	total after update
First	0	4
Second	1	7
Third	2	10
Fourth	3	13

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Move print inside the original **loop**. What sequence is displayed?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: **Trace** recommender **loop**s with current item, eligibility, score and current best so each update can be explained.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Trace recommender **loop**s with current item, eligibility, score and current best so each update can be explained.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional reading-token jar starts with one token and gains three per round.
- Which labelled step matches this detail? Print 13 once, after all four updates.
- 1. **Iteration** values
- 2. Updated totals
- 3. After the **loop**
- 4. Initial **state**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: After the **loop** Print 13 once, after all four updates. The turn value labels an **iteration**; it is not the number added. The fixed update adds three each time.

Four **iteration**s take the total from 1 to 13. Keeping print outside the **loop** produces one final display rather than four intermediate displays. Initialise total before the **loop** so a zero-**iteration** run still has a defined value.

Changed-case question

- Changed-case practice: Use range(0) instead of range(4). Which conclusion is supported by the supplied facts?
- 1. The **loop** body never runs and the printed total is 1.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The **loop** body never runs and the printed total is 1. The iterable contains no turn values; the initial assignment still happened.

Repair and save your work

Review Branching Logic (ai-2/3). Goal: **Trace** the selected branch in an if elif else chain. Short practice: Give expected labels for –1,0,10,11 and identify the equality boundary for Short..

Return to this week and explain what you changed.

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If print(total) moves before total=total+2, what gets printed for the original **input**?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a **trace** rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Trace Tables: the case

AI ACADEMY / AI-2/4 · 01 · LOOK AT THE FACTS

Trace Tables: the case

The situation

A fictional reading-token jar starts with one token and gains three per round.

Supplied evidence

Python: total=1; for turn in range(4): total=total+3; after the loop print(total). The update is inside the loop; print is outside.

By the end, I can make a trace table for a for-loop, including zero iterations.

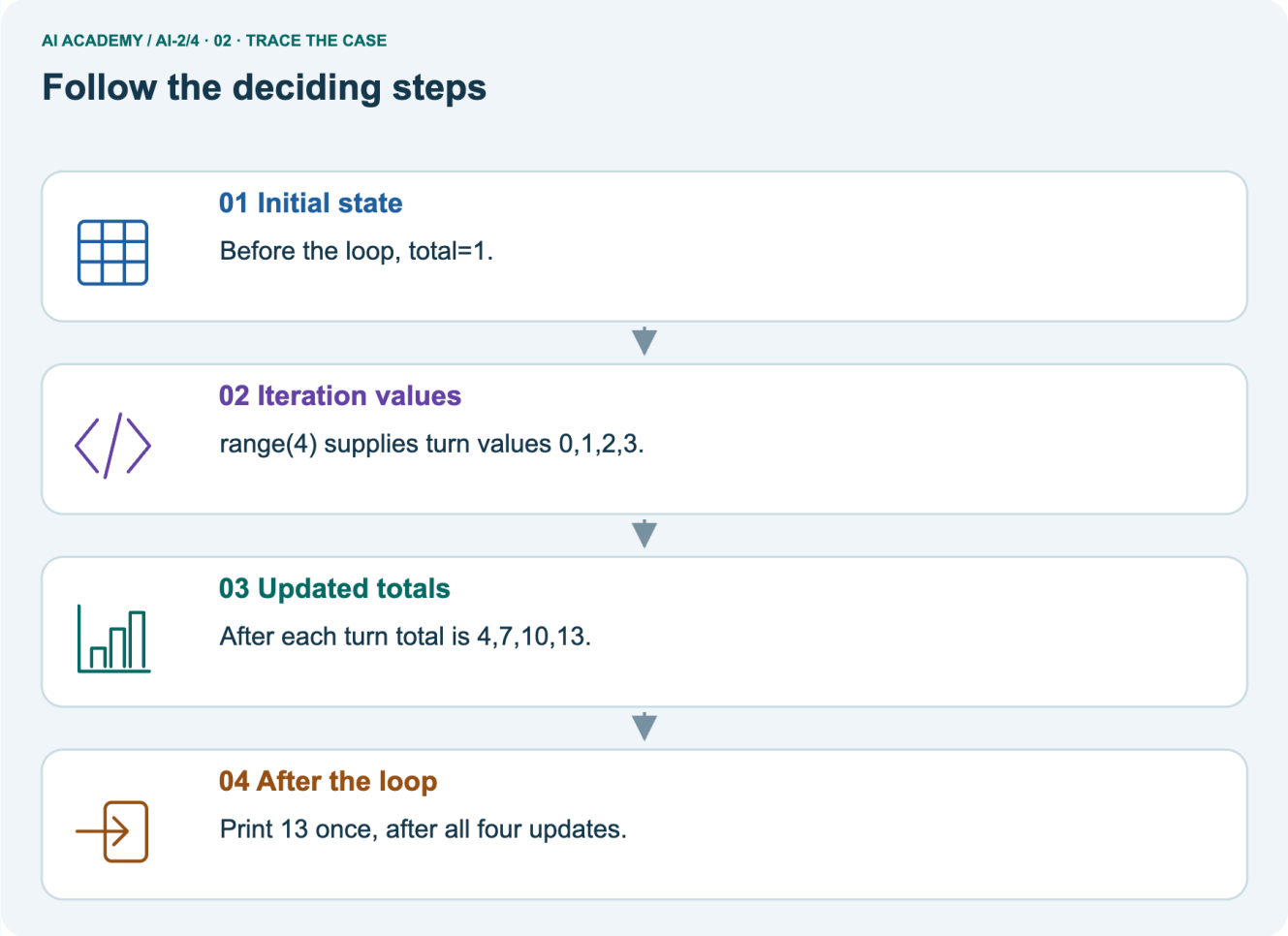
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional reading-token jar starts with one token and gains three per round. Supplied facts: Python: total=1; for turn in range(4): total=total+3; after the **loop** print(total). The update is inside the **loop**; print is outside.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



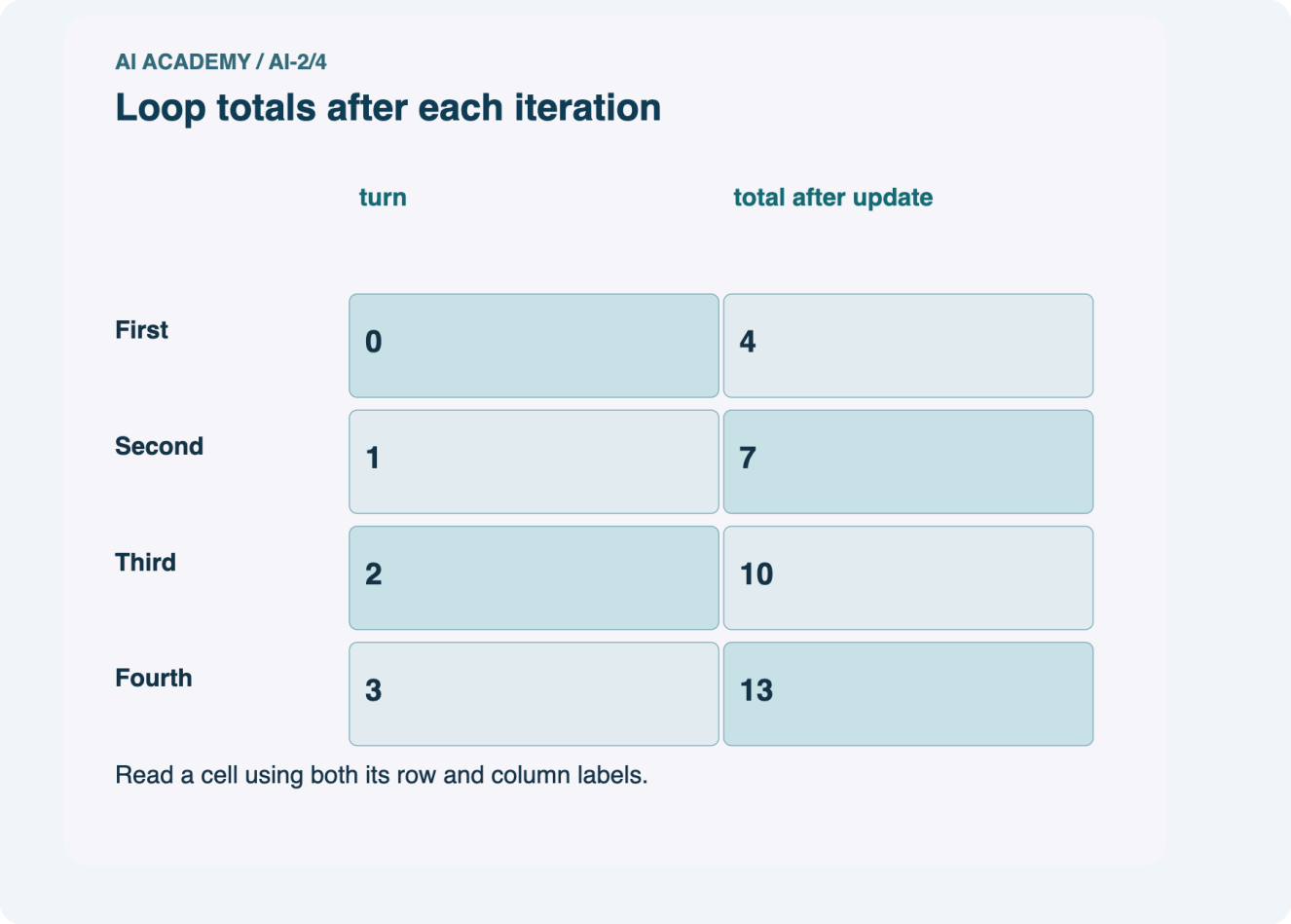
Follow the labelled picture with the matching explanation. Open the vector diagram

Initial **state**: Before the **loop**, total=1. → **Iteration** values: range(4) supplies turn values 0,1,2,3. → Updated totals: After each turn total is 4,7,10,13. → After the **loop**: Print 13 once, after all four updates.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Loop totals after each iteration



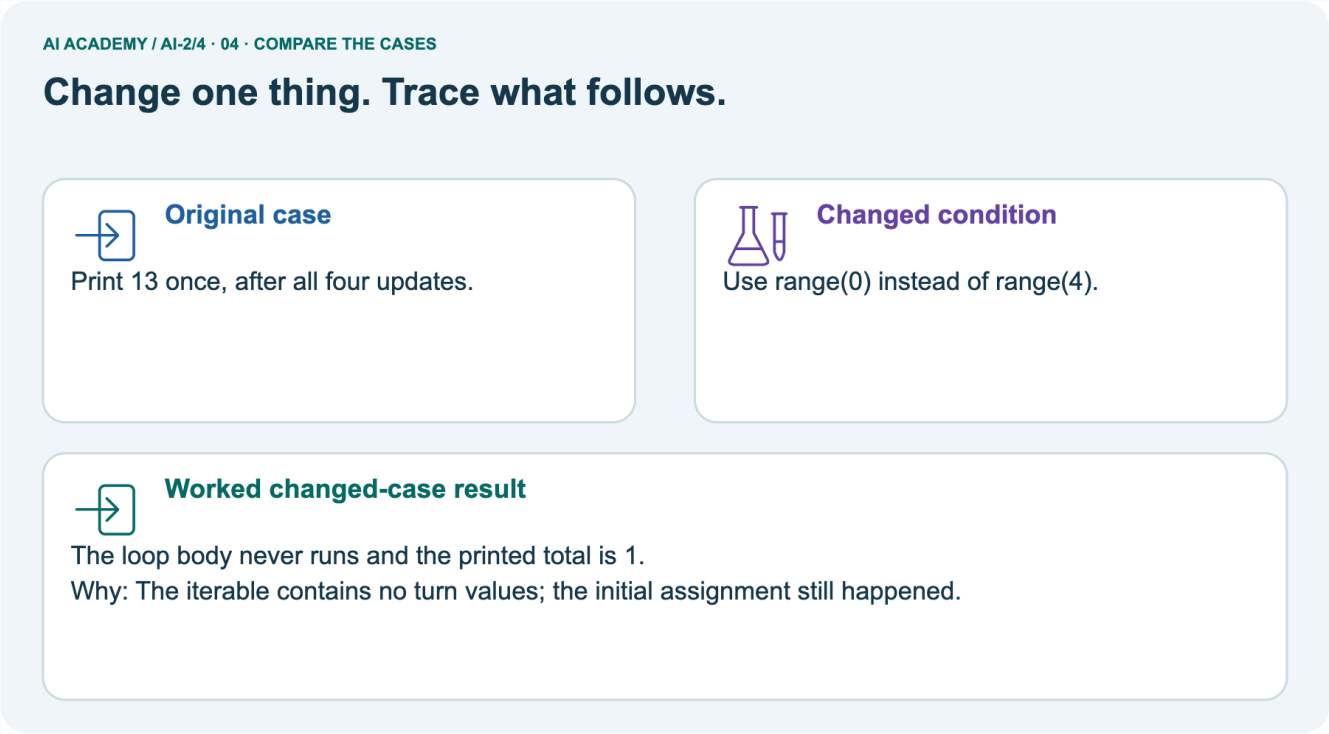
Follow the labelled picture with the matching explanation. Open the vector diagram

Loop totals after each **iteration**. First: turn = 0, total after update = 4. Second: turn = 1, total after update = 7. Third: turn = 2, total after update = 10. Fourth: turn = 3, total after update = 13.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

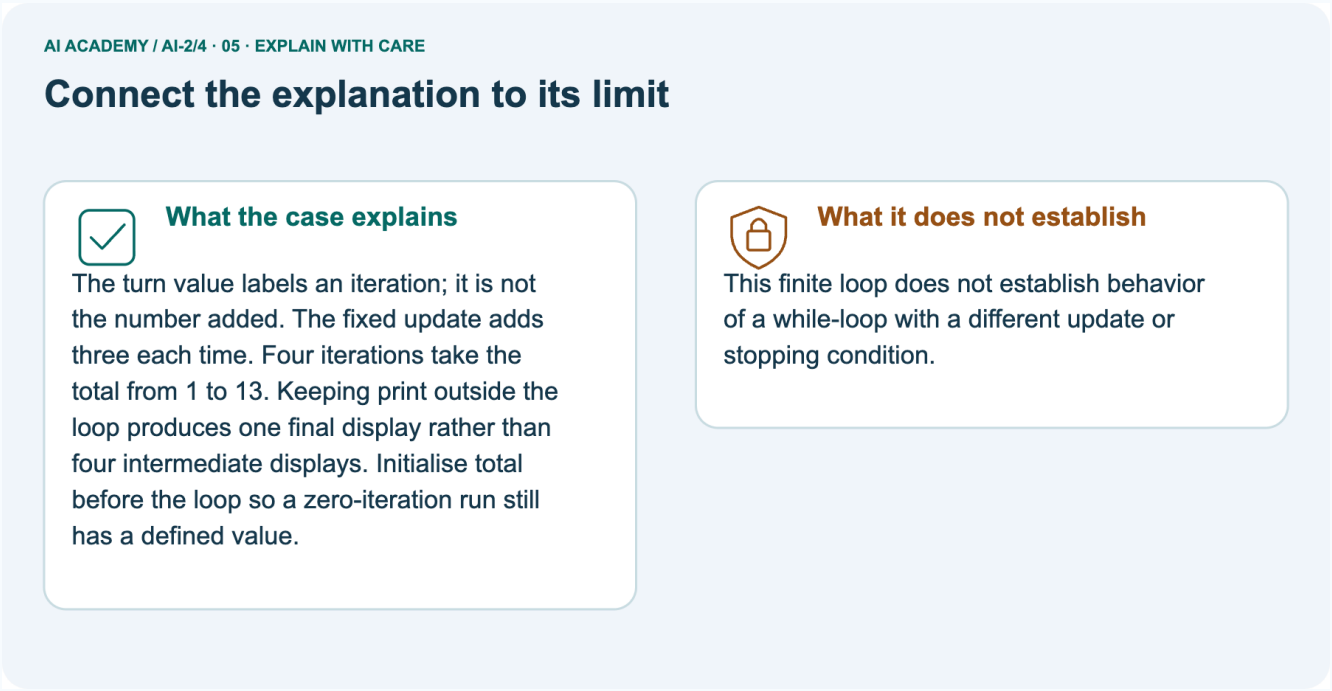
Original case: Print 13 once, after all four updates. Changed condition: Use range(0) instead of range(4). Expected result: The **loop** body never runs and the printed total is 1.

Reason: The iterable contains no turn values; the initial assignment still happened.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The turn value labels an **iteration**; it is not the number added. The fixed update adds three each time. Four **iteration**s take the total from 1 to 13.

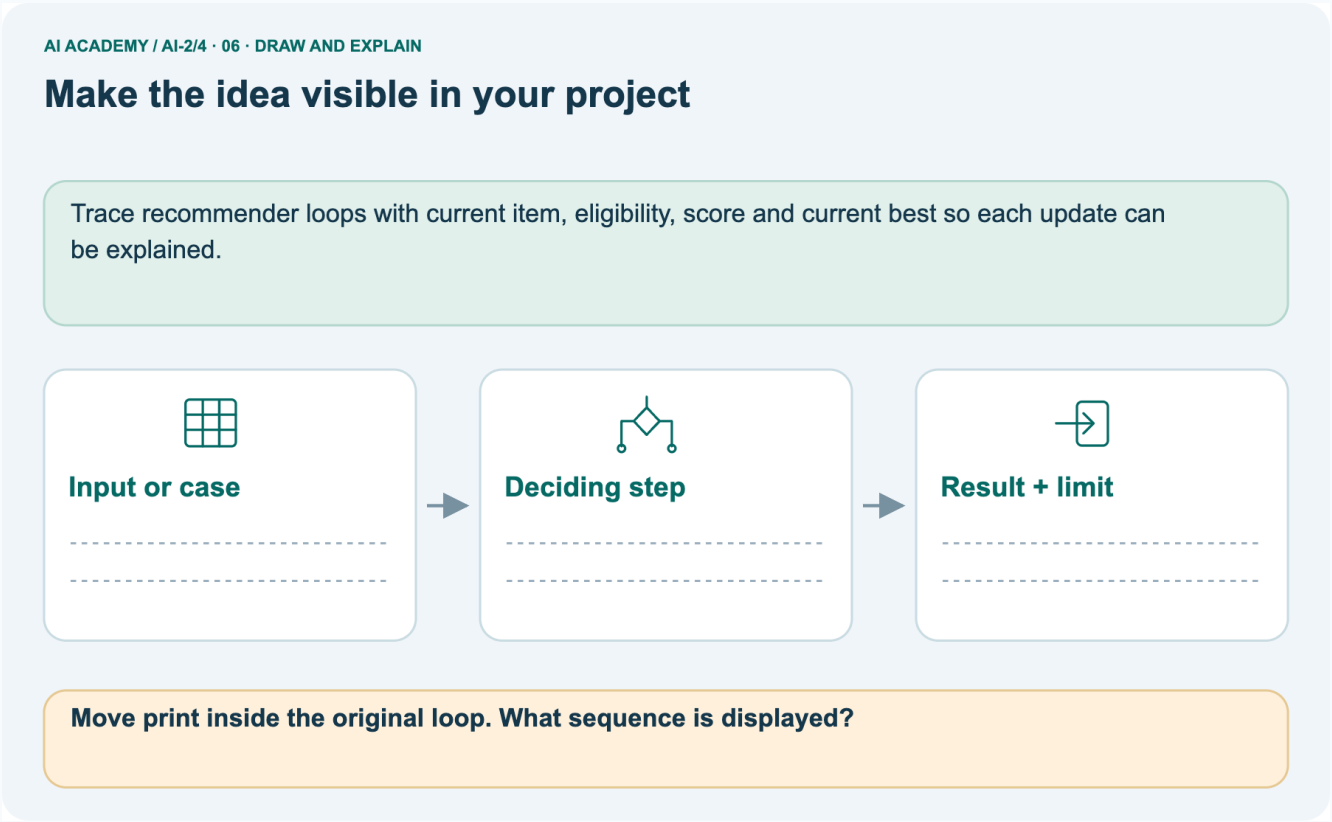
Keeping print outside the **loop** produces one final display rather than four intermediate displays. Initialise total before the **loop** so a zero-**iteration** run still has a defined value.

Limit: This finite **loop** does not establish behavior of a while-**loop** with a different update or stopping condition.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace recommender **loop**s with current item, eligibility, score and current best so each update can be explained. Draw your **input**, deciding step, result and limit. Supported question: Move print inside the original **loop**.

What sequence is displayed?

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Trace Tables: the case

02. Follow the deciding steps

03. Loop totals after each iteration

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.